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**Roll-no - 50**
**Class - SE3/S3**
**Experiment no - 5**

Topic - Develop a web application to fetch real-time weather data for a city. Use AJAX & a public weather API to retrieve information. Display temperature, humidity, and weather conditions dynamically.

**HTML code -**

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <meta name="description" content="A dynamic, real-time weather
application built with AJAX and the Fetch API.">
    <title>Dynamic Weather App</title>
    <!-- Use Modern Google Fonts for a Premium Feel -->
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com"
crossorigin>
    <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500&famil
y=Outfit:wght@400;600;700&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="style.css">
</head>
<body>

    <div class="glass-container">
        <h2>Weather Insights</h2>

        <div class="input-container">
            <input type="text" id="city" placeholder="Enter city
name..." autocomplete="off">
        </div>

        <button onclick="getWeather()">
            <span id="btnText">Get Weather</span>
            <div id="loader" class="loader"></div>
        </button>

        <div id="resultWrapper" class="result-container">
```

```
            <div id="result">
                <!-- Weather results will be injected here using DOM
Manipulation -->
            </div>
        </div>
    </div>

    <script src="script.js"></script>
</body>
</html>
```

**CSS code -**

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Inter', sans-serif;
    min-height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    background: linear-gradient(135deg, #1e3c72 0%, #2a5298 100%,
#1e3c72);
    background-size: 200% 200%;
    animation: gradientBG 10s ease infinite;
    color: #ffffff;
}

@keyframes gradientBG {
    0% { background-position: 0% 50%; }
    50% { background-position: 100% 50%; }
    100% { background-position: 0% 50%; }
}

.glass-container {
    background: rgba(255, 255, 255, 0.1);
    backdrop-filter: blur(20px);
    -webkit-backdrop-filter: blur(20px);
    border: 1px solid rgba(255, 255, 255, 0.2);
```

```
    border-radius: 24px;
    padding: 40px;
    width: 380px;
    box-shadow: 0 8px 32px 0 rgba(0, 0, 0, 0.37);
    text-align: center;
    transition: transform 0.3s ease;
}

.glass-container:hover {
    transform: translateY(-5px);
}

h2 {
    font-family: 'Outfit', sans-serif;
    font-weight: 600;
    font-size: 28px;
    margin-bottom: 30px;
    letter-spacing: 1px;
}

.input-container {
    position: relative;
    margin-bottom: 20px;
}

input {
    width: 100%;
    padding: 15px 20px;
    border: none;
    border-radius: 12px;
    background: rgba(255, 255, 255, 0.15);
    color: #fff;
    font-size: 16px;
    outline: none;
    transition: all 0.3s ease;
    font-family: 'Inter', sans-serif;
}

input::placeholder {
    color: rgba(255, 255, 255, 0.6);
}

input:focus {
```

```
    background: rgba(255, 255, 255, 0.25);
    box-shadow: 0 0 15px rgba(255, 255, 255, 0.1);
}

button {
    width: 100%;
    padding: 15px;
    border: none;
    border-radius: 12px;
    background: #ffffff;
    color: #1e3c72;
    font-size: 16px;
    font-weight: 600;
    font-family: 'Outfit', sans-serif;
    cursor: pointer;
    transition: all 0.3s ease;
    box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);
    position: relative;
}

button:hover {
    background: #f0f0f0;
    transform: translateY(-2px);
    box-shadow: 0 6px 20px rgba(0, 0, 0, 0.3);
}

button:active {
    transform: translateY(1px);
}

.result-container {
    margin-top: 30px;
    min-height: 120px;
    display: flex;
    flex-direction: column;
    justify-content: center;
    opacity: 0;
    transition: opacity 0.5s ease;
}

.result-container.visible {
    opacity: 1;
}
```

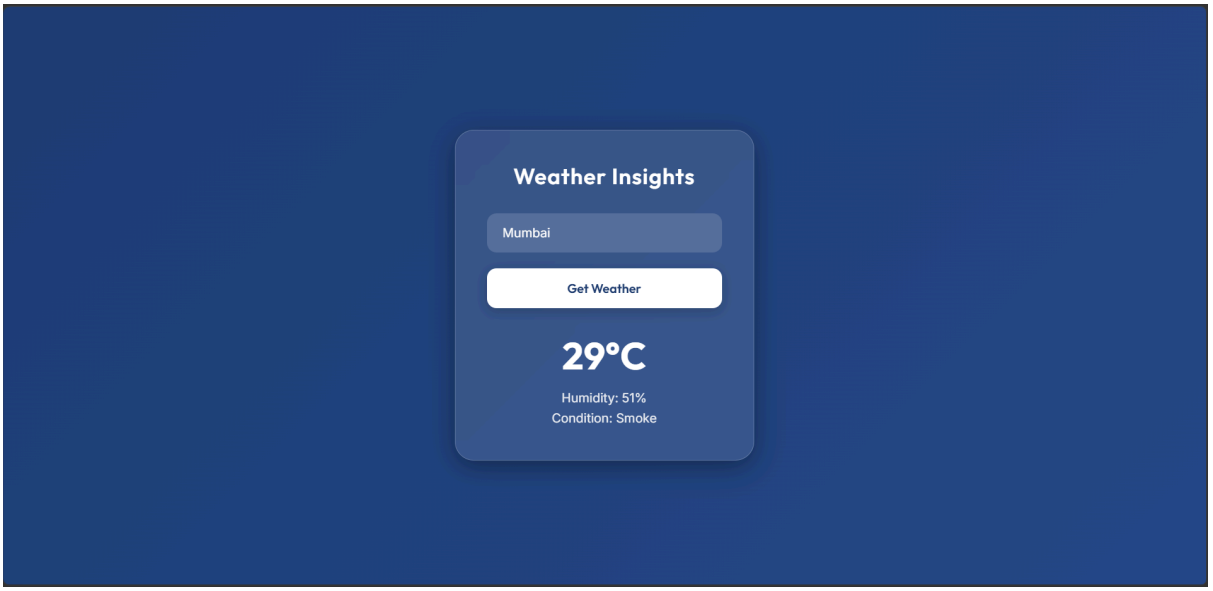
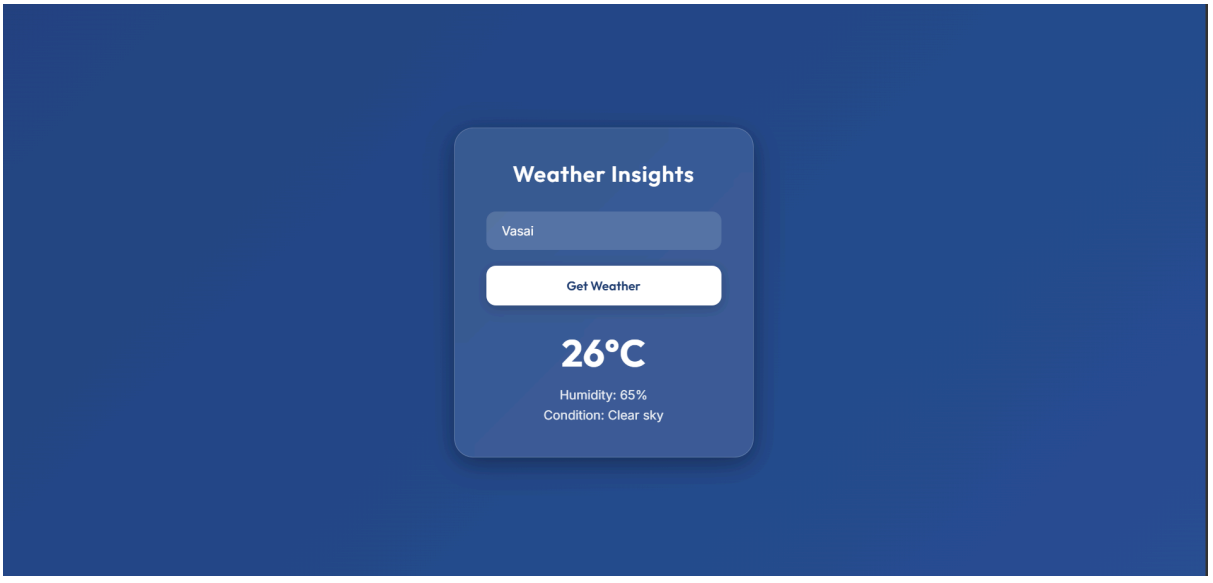
```
.temperature {
    font-size: 48px;
    font-weight: 700;
    font-family: 'Outfit', sans-serif;
    margin-bottom: 10px;
}

.details {
    font-size: 16px;
    color: rgba(255, 255, 255, 0.9);
    line-height: 1.6;
}

.error {
    color: #ff6b6b;
    font-weight: 500;
    font-size: 18px;
    line-height: 1.4;
    display: inline-block;
}

/* Loader */
.loader {
    border: 3px solid rgba(255, 255, 255, 0.3);
    border-radius: 50%;
    border-top: 3px solid #1e3c72;
    width: 20px;
    height: 20px;
    animation: spin 1s linear infinite;
    margin: 0 auto;
    display: none;
}

@keyframes spin {
    0% { transform: rotate(0deg); }
    100% { transform: rotate(360deg); }
}
```

**JavaScript code -**

```
function getWeather() {
    var cityInput = document.getElementById("city");
```

```
    var city = cityInput.value.trim();
    var resultContainer = document.getElementById("resultWrapper");
    var resultDiv = document.getElementById("result");
    var loader = document.getElementById("loader");
    var btnText = document.getElementById("btnText");

    // IMPORTANT!!!
    // Replace this with your actual OpenWeatherMap API Key
    var apiKey = "63e50a85c6a9df2a0802a4bfa387f9b8";

    if (!city) {
        resultContainer.classList.add("visible");
        resultDiv.innerHTML = '<span class="error">Please enter a city
name</span>';
        return;
    }

    if (!apiKey || apiKey === "YOUR_API_KEY") {
        resultContainer.classList.add("visible");
        resultDiv.innerHTML = '<span class="error">API key is
missing!<br>Please add your OpenWeatherMap API key in
script.js</span>';
        return;
    }

    var url = "https://api.openweathermap.org/data/2.5/weather?q=" +
encodeURIComponent(city) + "&units=metric&appid=" + apiKey;

    // Show loading state
    btnText.style.display = "none";
    loader.style.display = "block";
    resultContainer.classList.remove("visible");

    fetch(url)
        .then(response => {
            return response.json().then(data => {
                if (!response.ok) {
                    throw new Error(data.message || "City not found");
                }
                return data;
            });
        })
        .then(data => {
```

```
            // Processing JSON data
            var temp = Math.round(data.main.temp);
            var humidity = data.main.humidity;
            var description = data.weather[0].description;

            // Capitalize first letter of description
            description = description.charAt(0).toUpperCase() +
description.slice(1);

            // DOM Manipulation
            resultDiv.innerHTML = `
                <div class="temperature">${temp}°C</div>
                <div class="details">
                    <div>Humidity: ${humidity}%</div>
                    <div>Condition: ${description}</div>
                </div>
            `;
            resultContainer.classList.add("visible");
        })
        .catch(error => {
            // Error Handling
            var errorMsg = error.message;
            if (errorMsg) {
                // Capitalize first letter for display
                errorMsg = errorMsg.charAt(0).toUpperCase() +
errorMsg.slice(1);
            } else {
                errorMsg = "Something went wrong!";
            }
            resultDiv.innerHTML = `<span
class="error">${errorMsg}</span>`;
            resultContainer.classList.add("visible");
        })
        .finally(() => {
            // Restore button state
            btnText.style.display = "inline";
            loader.style.display = "none";
        });
}

// Add event listener to input field to trigger search on "Enter" key
press
```

```
document.getElementById('city').addEventListener('keypress', function
(e) {
    if (e.key === 'Enter') {
        getWeather();
    }
});
```

Output -